

SEXUALLY TRANSMITTED DISEASE

**SCE
DERMATOLOGY**

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**/ EBDV BY DR
EMAN GOMAA**



- CONNECT

SEXUALLY TRANSMITTED INFECTIONS

- SYPHILIS
- GONORRHEA
- CHANCROID
- LYMPHOGRANULOMA VENEREUM
- DONOVANOSIS (GRANULOMA INGUINALE)

SEXUALLY TRANSMITTED AND TRANSMISSIBLE PATHOGENS

Bacteria

Neisseria gonorrhoeae
Treponema pallidum
Haemophilus ducreyi
Chlamydia trachomatis
Mycoplasma hominis, *M. genitalium*
Ureaplasma urealyticum
Gardnerella vaginalis
Atopobium vaginae
Mobiluncus curtisii, *M. mulieris*
Klebsiella (Calymmatobacterium) granulomatis
Shigella spp.
Campylobacter spp.
Helicobacter cinaedi, *H. fennelliae*

Viruses

Human immunodeficiency virus, types 1 and 2
 Herpes simplex virus, types 2 > 1
 Human papillomavirus (several types)
 Hepatitis viruses, B > C and (via fecal-oral contact) A
 Cytomegalovirus
 Molluscum contagiosum virus
 Human T-cell leukemia/lymphotrophic virus, types I and II
 Human herpesvirus, type 8

Protozoa

Trichomonas vaginalis
Entamoeba histolytica
Giardia lamblia

Fungi

Candida albicans

Ectoparasites

Phthirus pubis
Sarcoptes scabiei

SYPHILIS

- SEXUALLY ACQUIRED, CHRONIC INFECTION CAUSED BY *TREPONEMA PALLIDUM* AND CHARACTERIZED BY A VARIETY OF CLINICAL MANIFESTATIONS AND INVOLVEMENT OF MULTIPLE ORGAN SYSTEMS

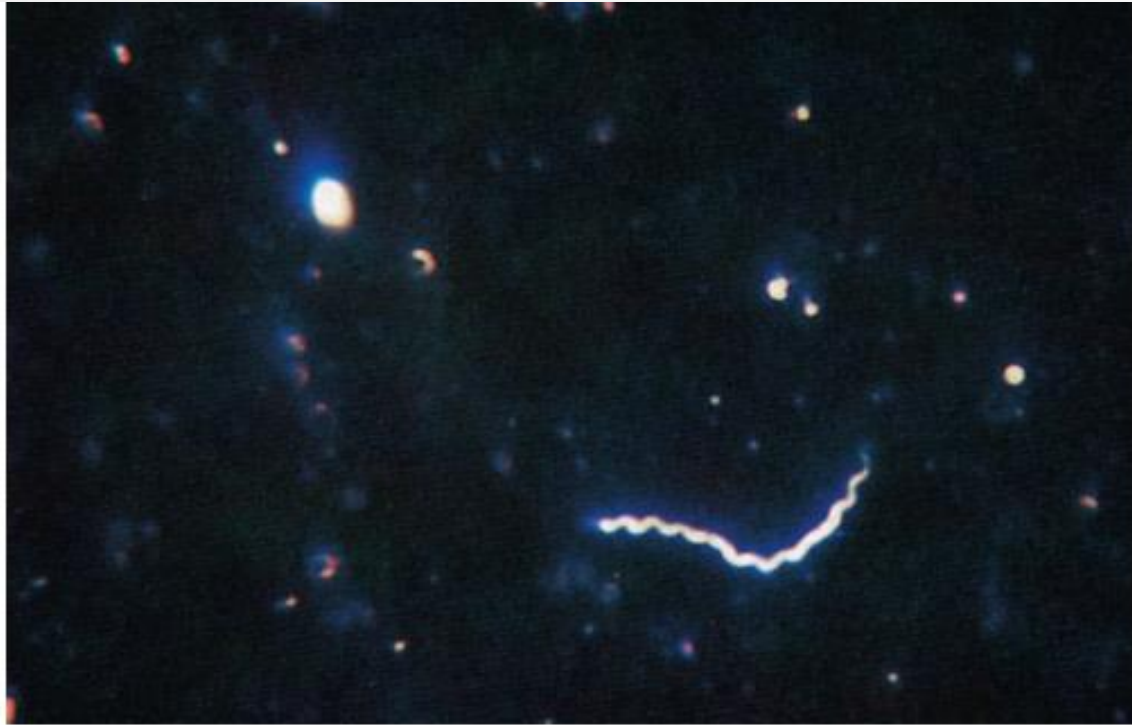


Fig. 82.3 Positive darkfield examination. Treponemes are recognized by their characteristic corkscrew shape, and deliberate forward and backward movement. (with permission from the copyright holder, Elsevier, for use in the book)

CLASSIFICATION OF *TREPONEMA* SPECIES

Order Spirochaetales

Genus *Leptospira*

Genus *Borrelia*

Genus *Treponema*

	Disease	Distribution	Transmission
<i>T. pallidum</i> subsp. <i>pallidum</i>	Venereal syphilis	Worldwide	Sexual; maternal-fetal
<i>T. pallidum</i> subsp. <i>pertenue</i>	Yaws	Tropics	Non-venereal (all ages)
<i>T. pallidum</i> subsp. <i>endemicum</i>	Endemic syphilis (Bejel)	Desert	Non-venereal (all ages)
<i>T. carateum</i>	Pinta	Tropics	Non-venereal (all ages)
<i>T. denticola</i> , <i>T.</i> <i>socranski</i>	Periodontal disease	Worldwide	Non-venereal

Common features shared by all spirochetes within the genus *Treponema*: morphology, antigenic properties, many DNA sequences, serologic detection, biochemical properties.

Variable features for spirochetes within the genus *Treponema*: transmission mode, geography, age of patients, appearance of clinical lesions.

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CHARACTERISTICS OF *TREPONEMA PALLIDUM*

- 6–20 microns in length, 0.1–0.18 microns in diameter
- Regular tight spirals
- Inability to survive outside an animal host
- Cannot be cultured *in vitro* for an extended time period
- Limited capacity for DNA repair
- Outer membrane:
 - lipid-rich
 - contains uniform-sized transmembrane rare outer membrane proteins (TROMP)
 - lacks lipopolysaccharide
- Periplasmic flagella

MODE OF INFECTION



Direct contact

Accidental

indirect

CLASSIFICATION

I. CONGENITAL:-

1. EARLY WITHOUT 1RY STAGE
2. LATE
3. STIGMATA

II. ACQUIRED → EARLY

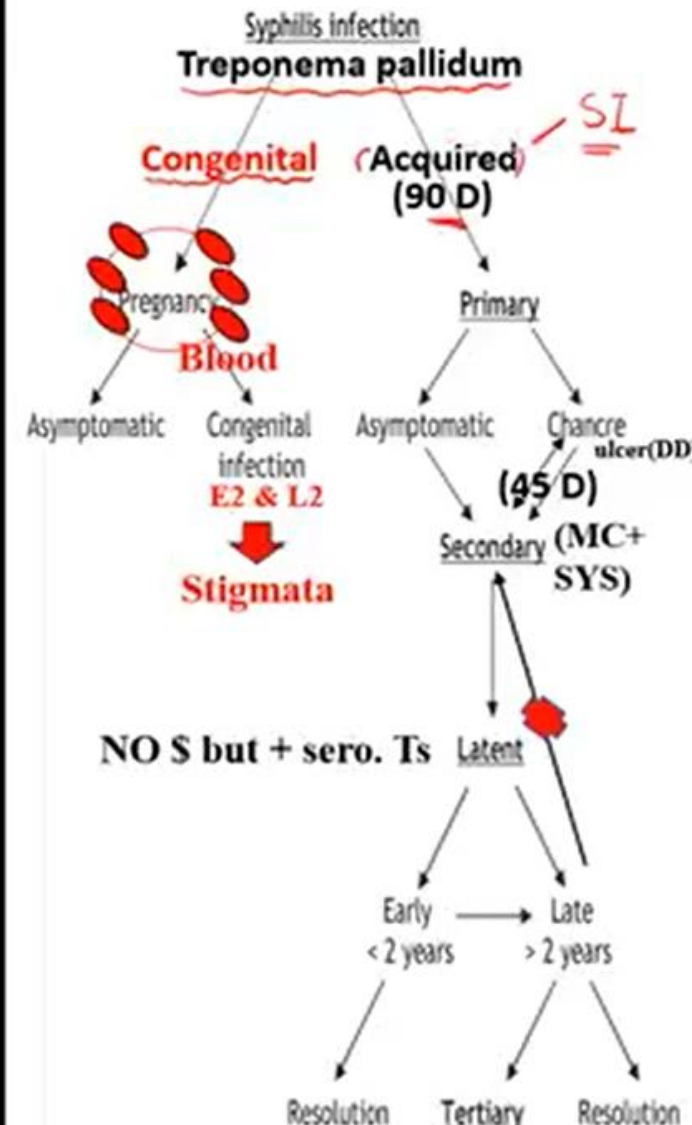
1. 1RY
2. 2RY
3. EARLY LATENT

→ LATE

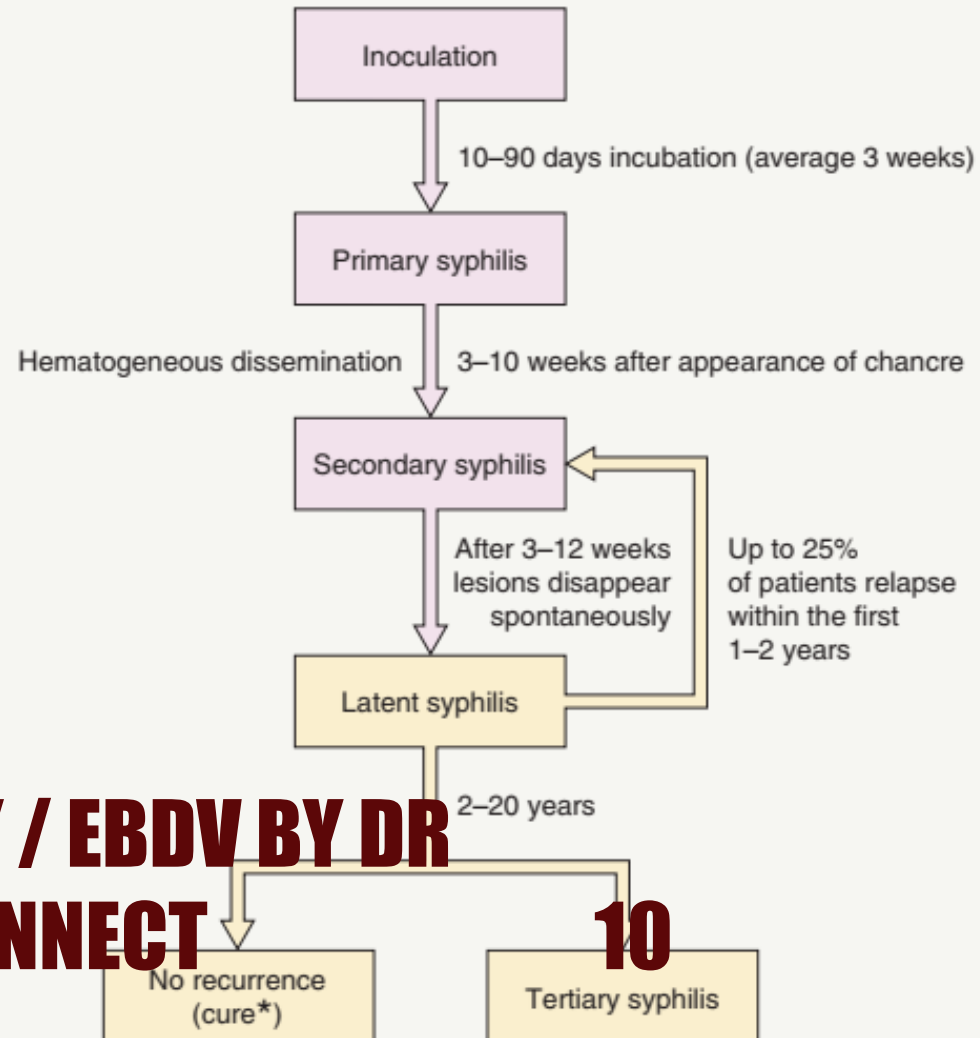
1. LATE LATENT
2. TERTIARY GAMMA
3. CVS - NEUROSYPHILIS

Syphilis

- **Organism:** Treponema pallidum
- **Mode of infection:**
 - ✓ **Acquired:**
 - ✓ **Sexual transmission:** MC
 - **Accidental:** direct (doctor), indirect (toilet seats) (very rare)
 - **Fresh blood transfusion:** if the donor is infectious.
 - **Congenital:**
 - **Vertical transmission:** mother baby.
- ✓ **Incubation period:** 9-90 days of infection
- ✓ **Clinical Picture:** →
- ✓ **Diagnosis:** SPECIFIC – NON S

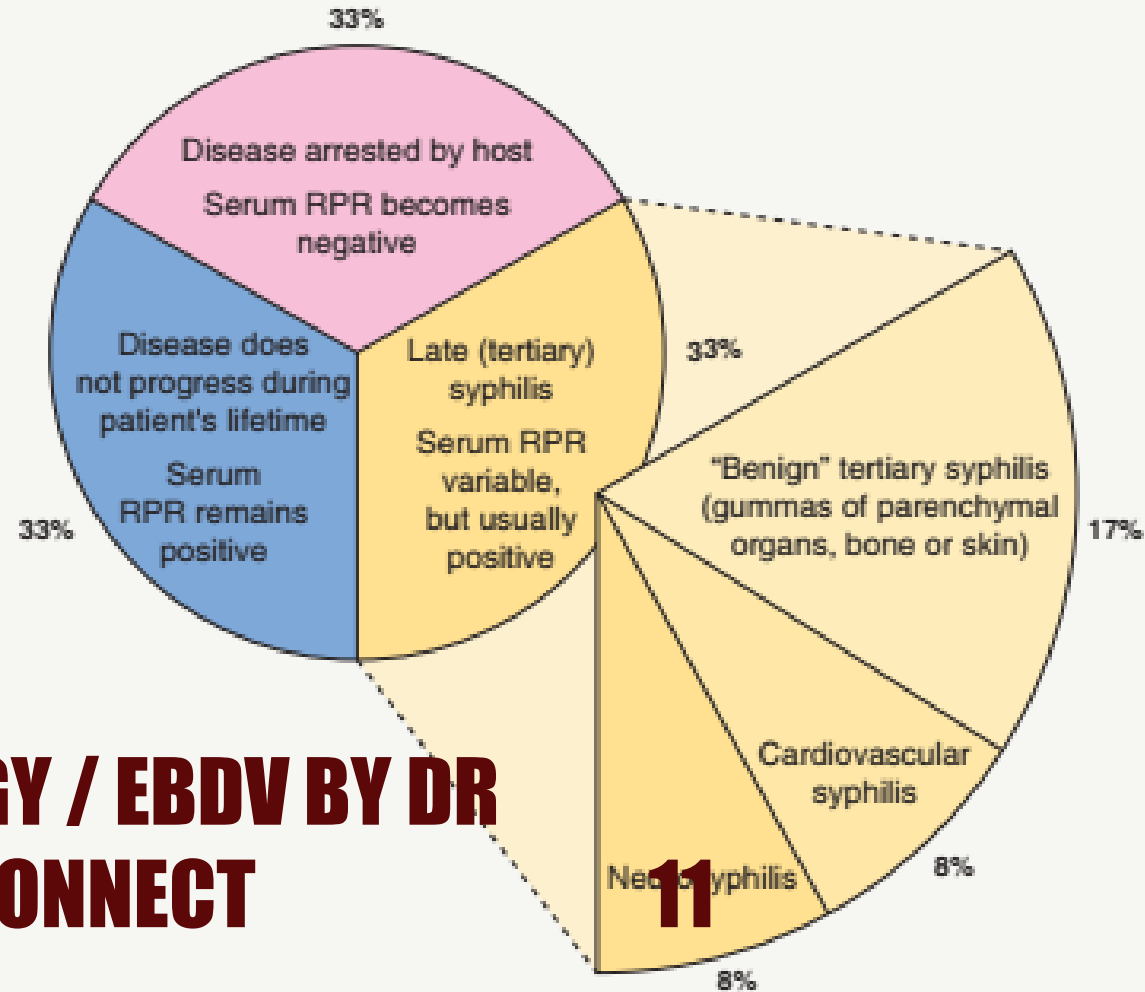


NATURAL HISTORY OF UNTREATED SYPHILIS



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NATURAL HISTORY OF UNTREATED SYPHILIS



1RY SYPHILIS [CHANCERE]

- AFTER IP OF 9-90 DAYS AT INOCULATION SITE.
- DULL RED PAPULE WHICH ULCERATE.
- 1. SINGLE BUT MAY BE MULTIPLE.
- 2. WELL DEFINED ROUNDED OR OVAL.
- 3. SLOPPING EDGE.
- 4. FLOOR → RED CLEAN GRANULATION TISSUE
- 5. INDURATED BASE OOZE SERUM
- 6. PAINLESS LN ENLARGEMENT → DISCRETE, RUBBERY WITH NORMAL OVERLYING SKIN.
- 7. GENITAL & EXTRAGENITAL.

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Primary stage (chancre)

Local reaction > treponemas > papule > syphilitic ulcer

- Num: Single.
- SITE : G-EXTRA G.
- Shape : round or oval
- Edge: sloping
- Floor: clean and fleshy red?
- Base: Indurated
- Pain: Painless if pain?
- LN: same + bilaterally
- Healing: 2-6 weeks - thin atrophic scar.



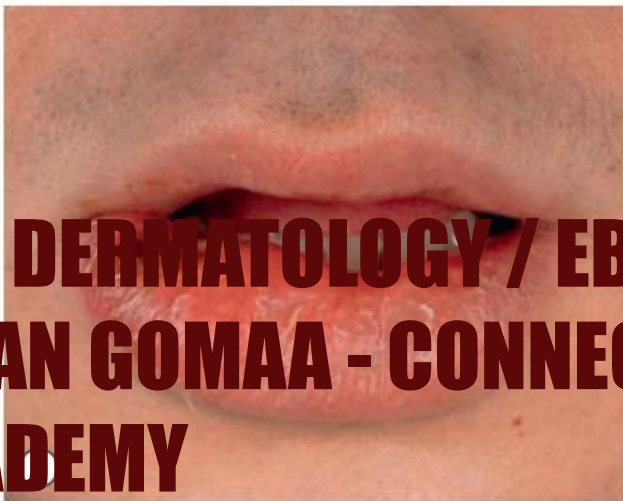


Fig. 82.5 Chancres of primary syphilis. The lesions are firm to palpation and are occasionally multiple. Sites of chancres can include the penis (**A**, **B**), perianal area (**C**) and lip (**D**). Occasionally, other sites are affected, e.g. fingers.

- **EXTRAGENITAL → LIPS, TONGUE, NIPPLE & ANUS.**
- **HEALING WITHIN 2-6 WEEKS WITH THIN ATROPHIC NON-CONTRACTILE SCAR.**

2RY SYPHILIS

- **6-12 WEEKS AFTER APPEARANCE OF CHANCER.**

AI SKIN RASH 75%

GENERALIZED, BILATERAL, SYMMETRICAL.

PALM & SOLE ARE COMMONLY AFFECTED

USUALLY NON-ITCHY

NEVER VESICULATE

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CLINICAL FEATURES OF SECONDARY SYPHILIS

- **Prodromal symptoms and signs:**
 - Weight loss
 - Low-grade fever
 - Malaise
 - Headache (meningeal irritation)
 - Sore throat
 - Conjunctivitis (iridocyclitis)
 - Arthralgia (periostitis)
 - Myalgias
 - Hepatosplenomegaly (mild hepatitis)
- **Generalized lymphadenopathy with indolent enlargement of lymph nodes (50–85%)**
- **Skin manifestations:**
 - Early (10%):* generalized eruption; non-pruritic, roseola-like, discrete macules, initially distributed on the flanks and shoulders
 - Late (70%):* generalized maculopapular and papulosquamous eruptions; more infiltrated lesions, often copper-colored; annular plaques on the face; corymbose arrangement (satellite papules around a larger central lesion); occurs in successive waves and is polymorphic
 - Localized syphilids* (specific infiltrations of treponemes; positive darkfield examination):
 - palms and soles: symmetric papules and plaques with a collarette of scale (collarette of Bielt)
 - anogenital area: condylomata lata
 - perioral area: "corona veneris" along the hairline
 - pigmented macules, mainly on the neck (postinflammatory; "necklace of Venus")
- **Manifestations involving mucous membranes (30%):**
 - Syphilitic perlèche, patchy papules
 - Mucous patches: "plaques muqueuses" in the oropharynx (equivalent to condylomata lata in the genital area)
 - Syphilitic sore throat: inflammation of the whole pharynx
- **Patchy alopecia (7%):** "moth-eaten" localized areas of hair loss; toxic

TYPES OF SKIN RASH :-

- **MACULAR SYPHILIDS → ROSE PINK DISCRETE MACULE ON TRUNK**
- **PAPULAR SYPHILIDS → DULL RED INDURATED PAPULES.**
- **PAPULO-SQUAMOUS SYPHILIDS.**
- **CONDYLOMA LATA → FLISHY, MOIST, SESSILE & DOESN'T BLEED EASILY → GENITALIA, PERI-ANAL, BUCCAL COMMISSURES → MOST INFECTIOUS LESIONS.**
- **PUSTULAR SYPHILIDS → IN IMMUNE COMPROMISED PATIENTS.**

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Fig. 82.7 Secondary syphilis. Widespread exanthem of pink papules (A), subtle minimally inflamed lesions localized to the arms in an HIV-positive man (B) and generalized papulosquamous lesions (C). Lesions on the palms (D) and soles (E, F) can have a collarette of scale. B, Courtesy, Kalman Watsky, MD.





Fig. 82.8 Orogenital lesions of secondary syphilis. Oral lesions can vary from small superficial ulcers (A) to mucous patches (B). Condylomata lata on the vulvar (C) and anal (D) areas may be misdiagnosed as HPV infection (i.e. condylomata acuminata).



Condyloma Acuminata vs Condyloma Latum

HPV

Syphi (2y)

- Rough verrucous
- Pedunculated
- Soft
- Bleeds easily
- Negative
- Negative



- Flat moist
- Sessile
- Indurated
- Does not bleed easily
- Positive
- Positive



- **BI MM LESIONS 33%**

MUCOUS PATCH → PAINLESS, ROUND, GREYISH WHITE INECROSIS]

LIPS, TONGUE, URETHRA & VAGINA

PLAQUE FOUCHES → DORSUM OF THE TONGUE → DUE TO DESTRUCTION OF FILIFORM PAPILLAE BY NECROTIC PROCESS

PATCH BECOME CONFLUENT → ULCERATE → SNAIL TRACK ULCER.

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Fig. 82.9 Less common manifestations of secondary syphilis. **A** Annular plaques with central hyperpigmentation on the forehead. **B** Split papule at the oral commissure. **C** Granulomatous nodules and plaques. **D** Necrotic lesion with scale-crust in a patient with AIDS. **E** "Malignant" syphilis with multiple necrotic, ulcerated and crusted lesions associated with severe constitutional symptoms. *D, Courtesy, Judith Stern, MD.*

- **C) GENERALIZED LYMPHADENOPATHY.**
- **D) GENERAL SYMPTOMS → FEVER, MALAISE, HAIR LOSS, UVEITIS, ARTHRITIS & PERIOSTITIS**

BENIGN LATE SYPHILIS [GAMMA SYPHILIS]

- **START 3-10 YS AFTER INFECTION → RARELY INFECTIOUS.**

- **AI SKIN:-**

ASYMMETRICAL, SOLITARY, PAINLESS.

DESTRUCTIVE WITH SLOW HEALING.

NO LN ENLARGEMENT.

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Fig. 82.11 Cutaneous gummas of tertiary syphilis. Arciform, erythematous eroded plaques with central scarring.

Latent syphilis (No \$ but positive sero.)

- **Spontaneous cure:**

Serological tests: negative and the disease "burnt out" (30% of cases).

- **Persistent latent Syphilis:**

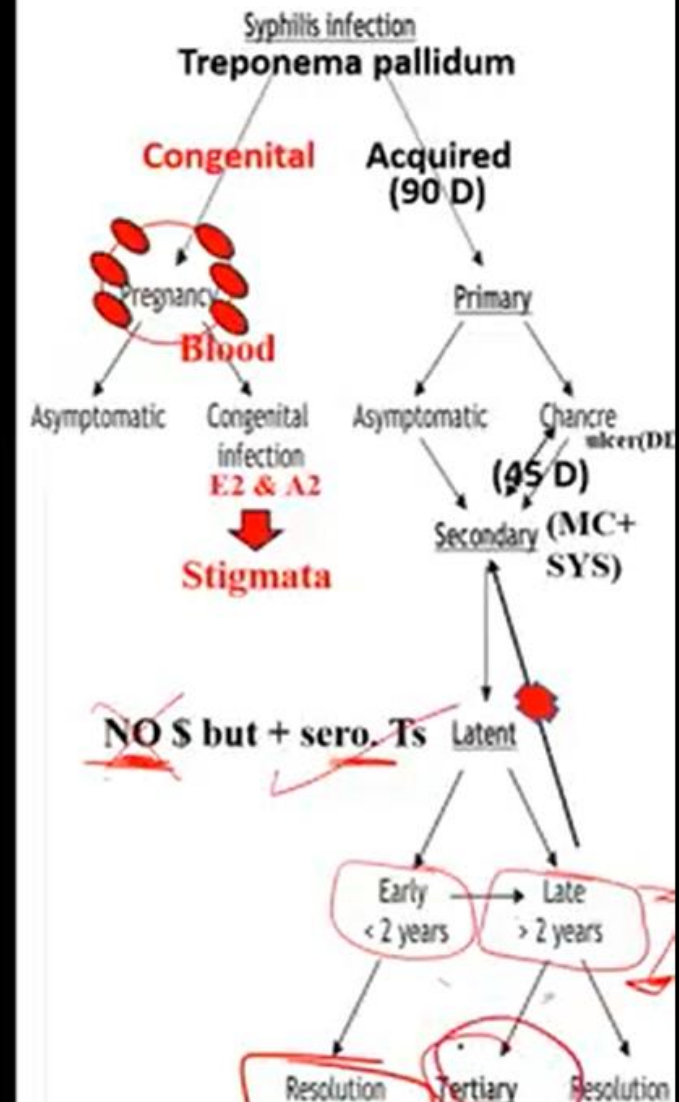
Serological tests : positive and no manifestations develop (30%).

- **Benign tertiary Syphilis:**

Gumma: in skin, mucous membranes, bones or internal organs (15%).

- **Chronic syphilitic inflammation of vital organs:**

manifestations of cardiovascular and neurosyphilis (25%).



SKIN

- 11 NODULAR.
- 21 PSORIASIFORM.
- 31 SC GAMMA → MOST COMMON → PAINLESS NODULE THAT ULCERATE.

ULCER → PUNCHED OUT BORDER, INDURATED BASE, NO LN, PAINLESS.

SITE → UPPER LEG, SCALP, FACE.

41 GAMMA OF MUCOUS MEMBRANE → DESTRUCTION OF UNDERLYING STRUCTURE → PERFORATED NASAL SEPTUM.

Tertiary stage (Gumma)

Allergic Response

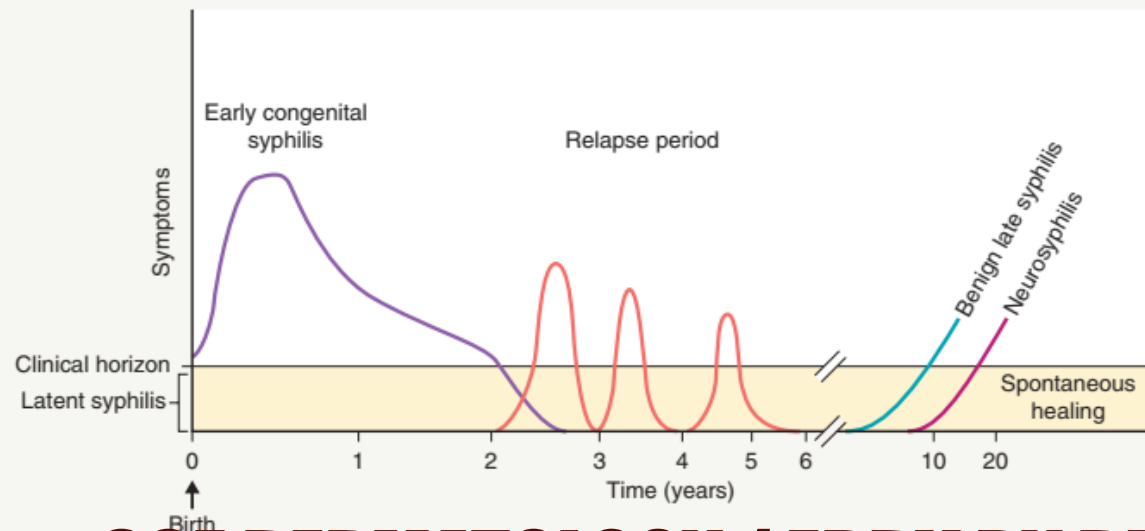
- Duration: 3-7 years of chancre.
- Gumma (Nodule or grouped nodules that ulcerate):
- Num: S&M
- Shape : polycyclic
- Site: MC + Any organs.
- Edge: punched out
- Floor: yellow necrotic material (Gummy)
- Base: Indurated
- Pain: Painless
- LN: NO LN ?



- **B) BONE & JOINT.**
- **C) VISCERAL SYPHILIS → LIVER (MOST COMMON), STOMACH LEATHER BOTTLE STOMACH**

CONGENITAL SYPHILIS

CLINICAL MANIFESTATIONS OF CONGENITAL SYPHILIS



MOTHER-TO-CHILD TRANSMISSION OF UNTREATED SYPHILIS AND ITS CONSEQUENCES

Risk

- Infection of the mother from conception to 7th month of pregnancy: risk of transmission is nearly 100% (often fetal demise or severe congenital syphilis)
- Infection at least 2 years before pregnancy: reduced risk of transmission to 50%
- Infection during 7th, 8th, or early 9th month: reduced risk of transmission
- Infection 3–6 weeks before labor: no placental transmission; risk of perinatal transmission

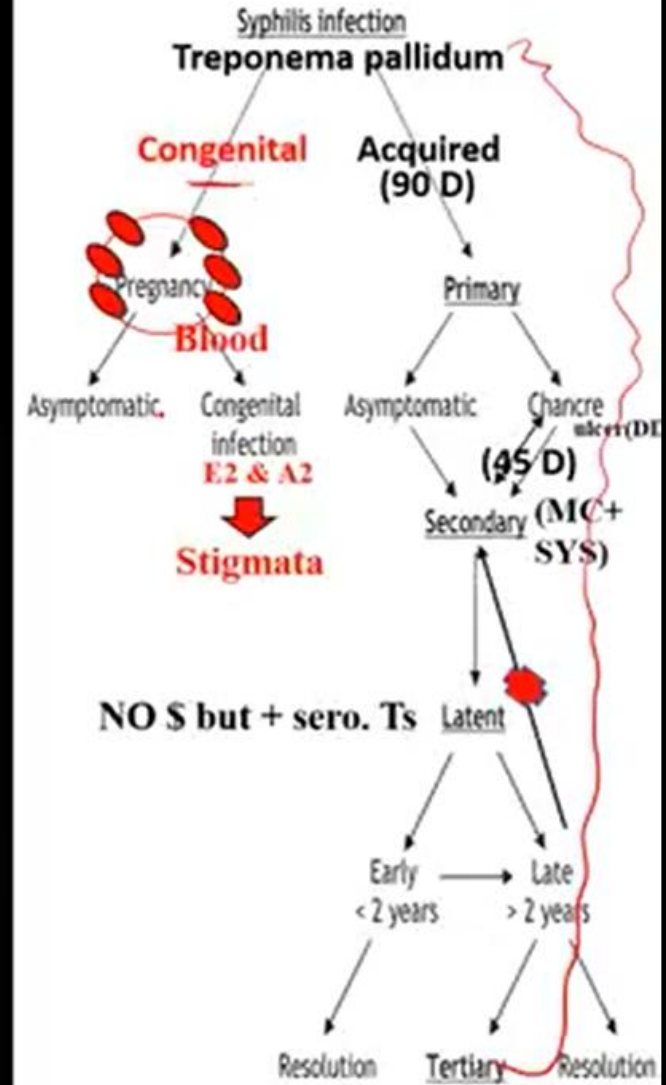
Consequences of infection

- Spontaneous abortion (second or third trimester) (10%)
- Stillbirth (10%)
- Infant death (20%)
- Congenital syphilis (20%)
- Healthy child (40%)

Congenital Syphilis

(TT>cross placenta>
No 1ry stage>No chancre)

- **Primary stage!:**
 - Syphilitic placenta is usually bulky, heavy, pale and greasy, infarcts are common?
- Escape infection.
- Miscarriage :12-16th week (placenta formation).
- Stillborn macerated baby.
- Born with syphilis \$.
- Born well: develops signs of **ECS** during 1st 2 years
- Born well: develops signs of **EC** after 2 years.





STIGMATA OF CONGENITAL SYPHILIS
Cutaneous
• Rhagades – radial periorificial (mouth, nose, eyes, anus) scars at sites of previous fissures
Dental
• Hutchinson teeth – peg-shaped, notched permanent incisors* • Mulberry molars – multiple rounded rudimentary cusps on the permanent first molars • Caries due to defective enamel
Skeletal
• Saddle nose – depression of the nasal root due to destruction of cartilage/bone • Frontal bossing of Parrot (“Olympian brow”) • Hypoplastic maxilla, relatively prominent mandible • High palatal arch ± perforation • Higouménakis’ sign – thickening of the medial clavicle • Scaphoid scapulae • Saber shins – anterior tibial bowing • Clutton’s joints – painless synovitis and effusions of the knees
Other
• Eighth nerve deafness* • Interstitial keratitis leading to corneal ulcers and opacities*
*Components of Hutchinson triad.

Early Congenital Syphilis(ECS)

(2ry S +Bone and joints)

- 1st 2 years.
- The skin rash:
 - Vesiculobullous eruptions (never occur in acquired syphilis).
 - Rhagades: around MM fissured and heal by radiating scars.
 - Purpuric eruption: thrombocytopenia.
- Bone and joint .

Infants with high titer of nasal discharge.

Late congenital syphilis

- 5 years.
- Frontal bossae -Saddle nose- Short maxilla.
- Rhagades
- Hutchinson's teeth
- High arched palate.
- Sabre tibia
- Clutton's joint.
- Paroxysmal cold hemoglobinuria
- Hutchinson's triad(pathognomonic)
 - Hutchinson's teeth
 - Interstitial keratitis
 - Saddle nose



DIAGNOSIS

- 1. HISTORY & EXAM.**
- 2. DARK FIELD MICROSCOPY → SPECIMEN (SEROUS FLUID FREE OF RBCS) → FINDING → MOTILE SPIROCHETES.**
- 3. PCR.**
- 4. SEROLOGICAL TEST**
REAGEN TEST (NON SPECIFIC)
SPECIFIC TEST USED TO CONFIRM THE NON-SPECIFIC ONE.

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Diagnosis of Syphilis

- ✓ Dark field microscopy *C1q*
- Syphilis Serology
 - **Non specific antibodies**
 - L. Specificity > false positive
 - H. Sensitivity > Screen – FU therapy.
 - VDRL & RPR
 - **Specific antibodies**
 - FTA-ABS IgM
 - TPHA
 - TPI



Specific antibodies

- Fluorescent Treponemal Antibody absorption (FTA-ABS IgM) test:

- First specific test to become positive
- IgM → Not cross placenta > can diagnose neonatal syphilis



- Treponema Pallidum Immobilization (TPI) test:

- Most specific

- Treponema Pallidum haemagglutination (TPHA) test:

- **5] PATHOLOGY**

1RY SYPHILIS → ULCERATION, DIFFUSE DERMAL INFILTRATE & SPIROCHETES.

2RY SYPHILIS → VARIABLE & SPIROCHETES.

3RY SYPHILIS → TUBERCLOID GAMMA + PLASM CELLS & DIFFICULT TO IDENTIFY SPIROCHETES

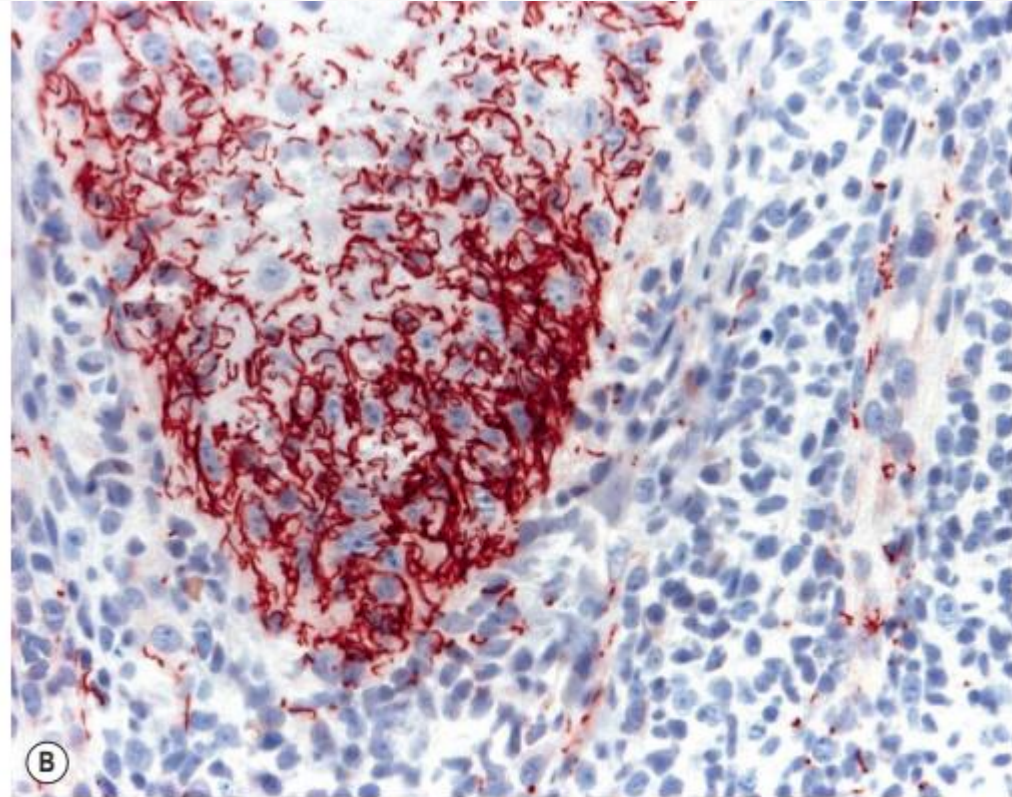
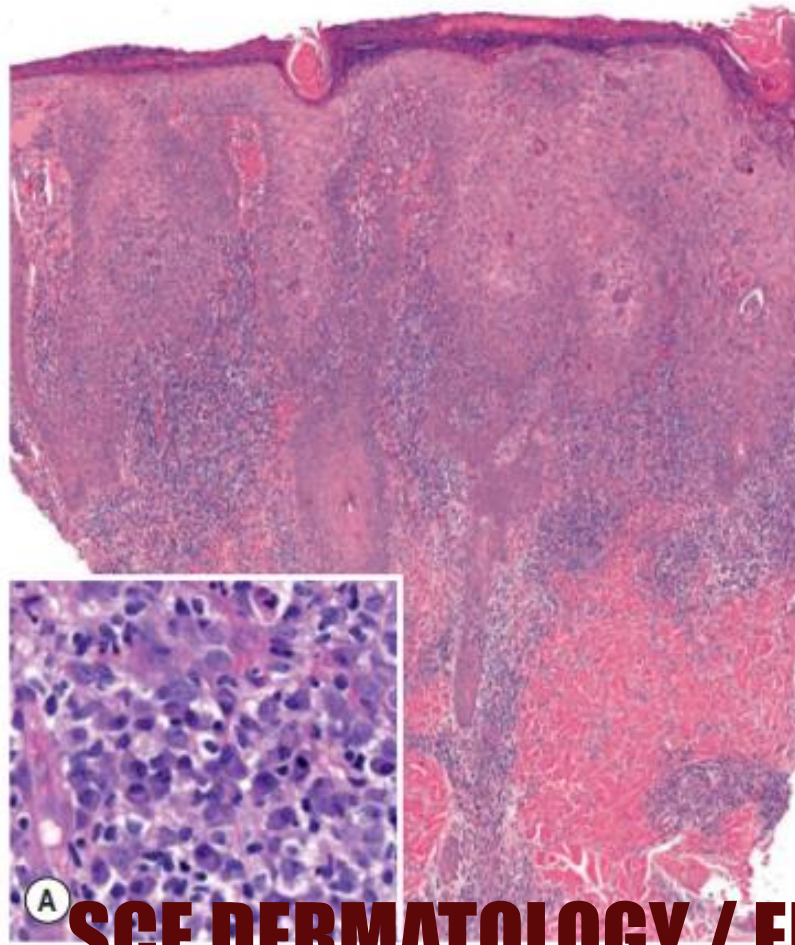


Fig. 82.15 Histology of secondary syphilis. **A** Psoriasiform epidermal hyperplasia and a dense dermal infiltrate of plasma cells (see inset), lymphocytes, and histiocytes are observed. **B** Positive immunohistochemical staining with an anti-*Treponema pallidum* antibody; note the characteristic

TTT

- **ACQUIRED:- BENZATHINE PENICILLIN 2.4 IU IM**

IF EARLY SINGLE DOSE.

LATE SYPHILIS → WEEKLY FOR 3 WEEKS.

PERINATAL SYPHILIS → BENZATHINE PENICILLIN.

PREGNANT → CEPHALOSPORINS

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Treatment of Syphilis

Penicillins

- Penzathine penicillin
 - Herxheimer reaction (therapeutic shock).
 - Therapeutic paradox.
 - Penicillin allergy.
- Erythromycin
- Tetracycline

TREATMENT RECOMMENDATIONS FOR SYPHILIS
Early syphilis (primary, secondary, and early latent [acquired <1 year previously])
<i>Recommended:</i>
• Benzathine penicillin, 2.4 million units* im as a single dose – or – • Procaine penicillin, 1.2 million units im daily for 10 days†
<i>Alternative regimens for penicillin-allergic patients‡:</i>
• Doxycycline, 200 mg daily (100 mg po BID preferred over a single 200 mg dose) for 14 days – or – • Tetracycline, 500 mg po four times daily for 14 days – or – • Ceftriaxone, 1–2 g im or iv daily for 10–14 days – or – • Azithromycin, 2 g po as a single dose§
Late latent (acquired >1 year previously or of unknown duration), cardiovascular and gummatous syphilis; retreatment of primary, secondary or latent syphilis after failed initial treatment¶
<i>Recommended:</i>
• Benzathine penicillin, 2.4 million units* im weekly for three doses – or – • Procaine penicillin, 1.2 million units im daily for 20 days†
<i>Alternative regimens for penicillin-allergic patients‡:</i>
• Doxycycline, 200 mg daily (100 mg po BID preferred over a single 200 mg dose) for 28 days – or – • Tetracycline, 500 mg po four times daily for 28 days
Neurosyphilis and ocular syphilis
<i>Recommended:</i>
• Aqueous penicillin G, 3–4 million units iv q4h (18–24 million units daily) for 10–14 days – or – • Procaine penicillin, 2.4 million units im daily plus probenecid, 500 mg po four times daily for 10–14 days

DIFFERENTIAL DIAGNOSES FOR SYPHILIS

Primary syphilis

Other causes of genital ulcers should be considered:

- Genital herpes
- Genital trauma
- Fixed drug eruption
- Ulcerative genital carcinoma (e.g. squamous cell carcinoma)
- Chancroid: jagged border, yellow exudate, painful
- Lymphogranuloma venereum: transient, indurated, painless
- Primary EBV infection
- Behçet disease

Secondary syphilis

- Cutaneous: pityriasis rosea, guttate psoriasis, viral exanthems, lichen planus, pityriasis lichenoides chronica, primary HIV infection, drug eruption, nummular eczema, folliculitis
- Mucous membranes: lichen planus; chronic aphthae; hand, foot and mouth disease; herpangina; perlèche
- Condylomata lata: warts due to HPV, bowenoid papulosis, squamous cell carcinoma

Tertiary syphilis

- Cutaneous: lupus vulgaris, chromoblastomycosis, dimorphic fungal infections, leishmaniasis, lupus erythematosus, mycosis fungoides, sarcoidosis, tumor, gumma, syphilitic ulcer

OTHER SEXUALLY TRANSMITED DISEASES

LYMPHOGRANULOMA VENEREUM

(LN) (LM4)

- Organism

- Chlamydia Trachomatis L-1,2,3.

- Incubation period:

- 2-3 weeks

- Clinical Picture:

- Multiple adenitis
- >matted and suppurate
- >multiple fused abscesses (buboes).
- >multiple sinus formation



Seek



Fig. 12.13 Lymphogranuloma venereum (LGV). Inguinal bubo that has ruptured and

CLINICAL MANIFESTATIONS OF LYMPHOGRANULOMA VENEREUM

Initial manifestations: 3–12 days

- Papule
- Erosion or ulcer
- Herpetiform vesicle
- Nonspecific urethritis or cervicitis

Inguinal syndrome: 10–30 days up to 6 months

- Regional lymphadenopathy (mostly inguinal and femoral; also perirectal, deep iliac)
- Overlying erythema
- Constitutional symptoms
- Eruption of buboes
- Pelvic inflammatory disease (PID), back pain

Ano-genito-rectal syndrome: months to years

- Proctocolitis
- Hyperplasia of intestinal and perirectal lymphatic tissue
- Perirectal abscesses
- Ischiorectal and rectovaginal fistulas
- Anal fistulas
- Rectal strictures and stenoses

Other manifestations

- Urethro-genito-perineal syndrome
- Perianal elephantiasis
- Erythema nodosum
- Submaxillary or cervical lymphadenopathy associated with

CHANCROID

CROI > CRY > PAIN

- Organism:

- Haemophilus ducreyi, gram-negative bacilli.

- Clinical Picture:

- M. Papule > pustular > ulcerated.
 - Pain: tender and quite painful.
 - Edges: ragged and undermined.
 - Base: granulation tissue with a bloody exudate on manipulation.
 - No unilateral lymphadenopathy.



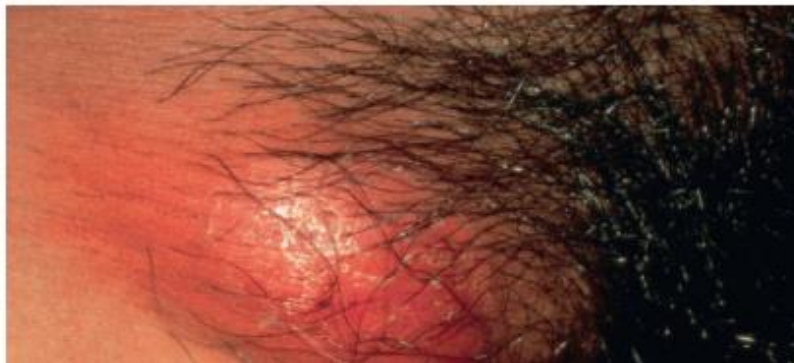


Fig. 82.22 Chancroid. **A** Well-demarcated painful ulcers on the penis. **B** Multiple purulent ulcers with undermined borders. **C** Unilateral lymphadenitis with overlying erythema. *B, Courtesy, Joyce Rico, MD*

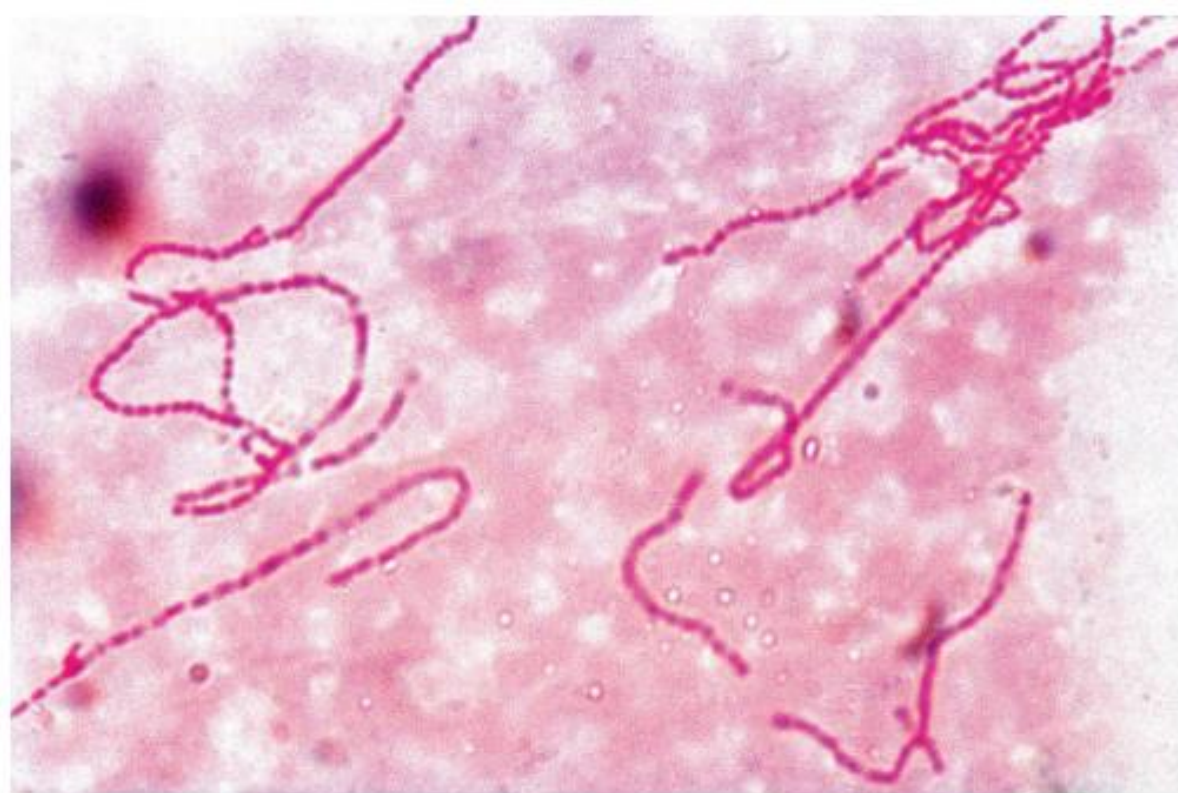


Fig. 82.21 Gram-stained smear in chancroid. This smear of exudate from a genital ulcer shows the characteristic chaining pattern of *Haemophilus ducreyi*.

	<u>Chancre</u>	<u>Chancroid</u>
<u>Incubation period</u>	9-90 days	4-10 days
<u>Number of lesions</u>	single	multiple
<u>Pain</u>	painless	painful
<u>Edge</u>	sloping	ragged undermined
<u>Base</u>	firm indurated	soft
<u>Bleeding</u>	does not bleed easily	bleeds easily
<u>Lymph nodes</u>	painless, discrete, firm, never suppurate	tender, matted, suppurate and form sinuses



TREATMENT REGIMENS FOR CHANCROID

- Azithromycin, 1 g po, single dose – *or* –
- Ceftriaxone, 250 mg im, single dose – *or* –
- Ciprofloxacin, 500 mg po twice daily for 3 days^{*,†} – *or* –
- Erythromycin base, 500 mg po four times daily for 7 days[†]

*Contraindicated for pregnant or lactating women.

†Worldwide, isolates with intermediate resistance.

GRANULOMA INGUINALE (KALEMAT)

- Organism:

- *Calymmatobacterium granulomatis*

- Clinical Picture:

- 1ry: Papules or nodules >ulcerate.
- LN: No > Granulomatous mimic lymph nodes (**pseudo-buboes**).

- Complications:

- Stenosis of the urethral, vaginal or anal orifices, and massive lymphatic edema





Fig. 82.25 Donovanosis (granuloma inguinale). Large ulcers with a characteristic "beefy" appearance. Courtesy, Joyce Rico, MD

GONORRHEA



Fig. 82.16 Gonococcal urethritis with a purulent urethral discharge.

CLINICAL MANIFESTATIONS OF GONORRHEA

Disseminated infection

- Arthritis
- Fever
- Tenosynovitis
- Acral cutaneous pustules
- Scalp abscesses (in neonates at sites of fetal scalp monitor electrodes)
- Endocarditis
- Meningitis

Direct mucosal infection

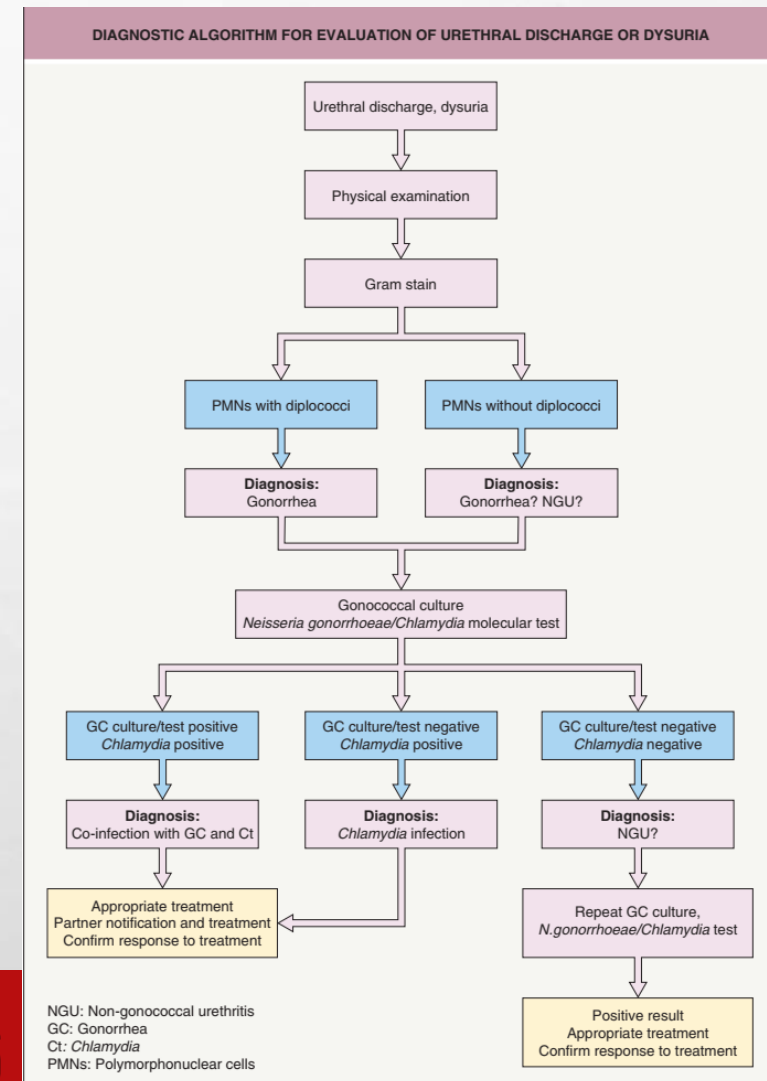
- Urethritis
- Cervicitis
- Proctitis
- Pharyngitis
- Vulvovaginitis (children)
- Ophthalmia neonatorum

Local extension

- Prostatitis
- Vesiculitis
- Epididymitis
- Salpingitis
- Oophoritis
- Pelvic inflammatory disease



Fig. 82.17 Gonococcemia (arthritis-dermatosis syndrome). Pustule with surrounding erythema on the toe.



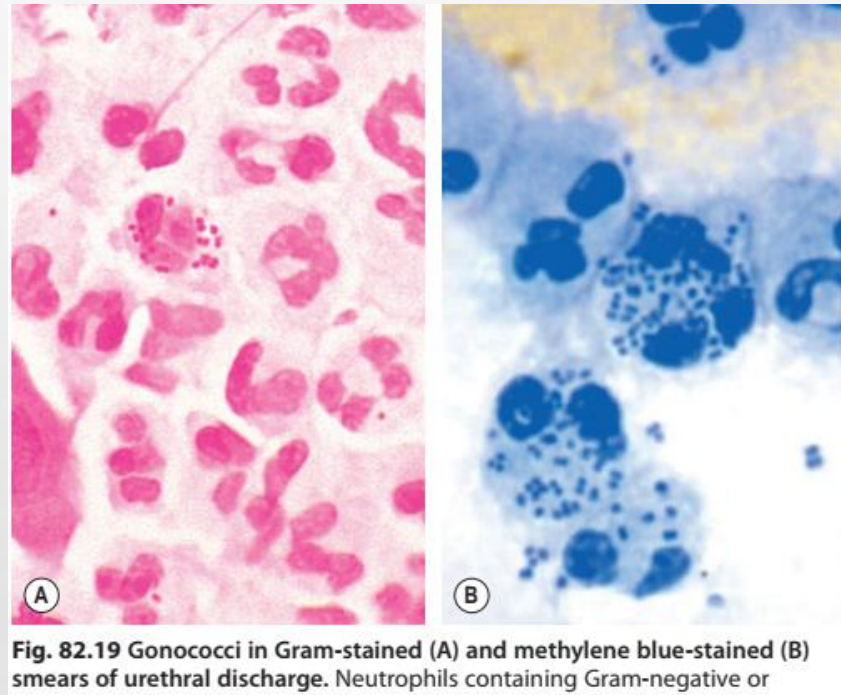


Fig. 82.19 Gonococci in Gram-stained (A) and methylene blue-stained (B) smears of urethral discharge. Neutrophils containing Gram-negative or

INFECTIOUS CAUSES OF GENITAL ULCER DISEASE				
Disease	Incubation time	Clinical lesion	Diagnosis	Organism
Genital herpes	3–7 days	Vesicles, erosions, ulcers; history of herpes viral infection; painful	PCR, culture, DFA, Tzanck (if vesicles)	HSV 2 > 1
Primary syphilis	10–90 days, average 3 weeks	Non-purulent; usually single ulcer; indurated; relatively painless	Darkfield microscopy, serology, PCR	<i>Treponema pallidum</i>
Chancroid	3–10 days	Purulent; often multiple ulcers; soft, undermined edges; painful	Culture, PCR	<i>Haemophilus ducreyi</i>
LGV	3–12 days	Transient ulcer; indurated; painless	PCR, culture, serology	<i>Chlamydia trachomatis</i> serovars L1–3
Donovanosis	2–12 weeks	Chronic ulcer; indurated, beefy red, friable	Smears, histology	<i>Klebsiella (Calymmatobacterium) granulomatis</i>

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THANK YOU

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**SCE DERMATOLOGY / EBDV BY DR
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